Guided Notes: Configuring Routing and Advanced Switching (Network+ N10-009 Module 5)

Module Objectives

- Compare and contrast routing concepts and dynamic routing.
- Install and troubleshoot routers.
- Explain tiered switching architecture.
- Explain virtual LANs (VLANs).

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1. Routing Technologies

Routing Tables and Path Selection

- Routing entries are defined by key parameters such as destination, next hop, metric, and interface.
- Routing tables store these entries to help routers decide where to forward packets.

Static Routes

- Manually added by administrators.
- Do not change unless manually edited.
- Best for small or stable networks.

Packet Forwarding Process

1. Receive the packet.
2. Examine the packet.
3. Consult the routing table.
4. Make a forwarding decision.
5. Forward the packet to the next hop.

Router Configuration Tools

- Various commands and utilities help configure and troubleshoot routers.

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2. Dynamic Routing Technologies

Static vs. Dynamic Routing Comparison

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic adaptation to network changes
Flexibility	Inflexible	Flexible, real-time updates
Control	Complete control over paths	Less direct control
Use Case	Small, stable networks	Large, complex, frequently changing networks

Common Dynamic Routing Protocols

- **RIP** (Routing Information Protocol): Legacy protocol finding best paths by hop count.
- **EIGRP** (Enhanced Interior Gateway Routing Protocol): Cisco proprietary, efficient.
- **OSPF** (Open Shortest Path First): Link-state protocol using Shortest Path First algorithm.
- **BGP** (Border Gateway Protocol): Used for routing between autonomous systems on the Internet.

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3. Network Address Translation (NAT)

Purpose

- Translates private IP addresses to public IP addresses for internet access.

NAT Types

- Static NAT, Dynamic NAT, and Port Address Translation (PAT).
- PAT allows multiple devices to share a single public IP using different ports.

Edge Routers

- Typically handle NAT and connect internal networks to external networks.

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4. Firewalls

Firewall Types

- Packet filtering, stateful inspection, proxy, and next-generation firewalls.

Stateless vs. Stateful Firewalls

- Stateless: Filters packets based on headers only.
- Stateful: Tracks state of active connections for more intelligent filtering.

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5. Enterprise Networking Topologies

Basic Network Topologies

- Bus, Star, and Ring.

Three-Tiered Network Hierarchy

- Core, Distribution, and Access layers for scalable and manageable networks.

Hybrid Topologies

- Combine basic topologies to enhance redundancy and fault tolerance.

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6. Virtual LANs (VLANs)

Purpose

- Create separate logical networks within a single physical network.

VLAN IDs and Membership

- VLANs are identified by IDs; ports can be members of one or more VLANs.

Tagged vs. Untagged Ports

- Tagged ports carry VLAN identifiers.
- Untagged ports carry traffic without VLAN tags.

VLAN Types and Trunking

- VLAN trunking bundles multiple VLANs over a single physical link using IEEE 802.1Q standard.

VLAN Routing

- Routers or Layer 3 switches route traffic between VLANs.
- "Router on a stick" uses a single router interface connected to a trunk port.

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7. Routing and VLAN Troubleshooting

Common Issues

- Routing issues: Can ping default gateway but not remote hosts.
- Default route and routing loops: Misconfigured default routes can cause loops.
- VLAN assignment: Devices must have matching IP configurations for their VLAN.

Summary: Key Concepts

- **Routing Tables:** Internal router maps directing packet forwarding.
- **Dynamic Routing:** Protocols automate route updates.
- **NAT:** Translates private IPs for internet access.
- **VLANs:** Logical segmentation of networks within physical infrastructure.

Vocabulary List

- **Routing Table:** A database in a router that stores routes to various network destinations.
- **Static Route:** A manually configured routing entry that does not change unless edited by an administrator.
- **Dynamic Routing:** A routing method where protocols automatically adjust routes based on network changes.
- **Packet Forwarding:** The process by which a router receives a packet, consults the routing table, and sends it toward its destination.
- **Routing Protocol:** Algorithms used by routers to share and update routing information.
- **Network Address Translation (NAT):** A method that translates private IP addresses to public IP addresses for internet communication.
- **Edge Router:** A router that connects an internal network to external networks, often handling NAT.
- **Port Address Translation (PAT):** A type of NAT that allows multiple devices to share a single public IP by using different ports.
- **Firewall:** A security device or software that monitors and controls incoming and outgoing network traffic.
- **Stateless Firewall:** A firewall that filters packets based only on header information without tracking connection state.
- **Stateful Firewall:** A firewall that tracks active connections and filters packets based on connection state.
- **Network Topology:** The arrangement or layout of devices and connections in a network.
- **Three-Tiered Network:** A hierarchical network design with core, distribution, and access layers.
- **Hybrid Topology:** A network topology combining two or more basic topologies to enhance redundancy and fault tolerance.
- **Virtual LAN (VLAN):** A logical network segment created within a physical network to separate traffic.
- **Tagged Port:** A switch port that carries VLAN information (tags) in data packets.
- **Untagged Port:** A switch port that carries data packets without VLAN tags.
- **Trunking:** The practice of combining multiple VLANs on a single physical link using tagging protocols.
- **Router on a Stick:** A router configuration with a single interface connected to a trunk port to route traffic between VLANs.
- **Routing Loop:** A network problem where packets circulate endlessly due to misconfigured routing.
- **Default Route:** A route used by a router when no specific route to a destination is found.

Acronyms List

- **NAT:** Network Address Translation
- **PAT:** Port Address Translation
- **RIP:** Routing Information Protocol
- **EIGRP:** Enhanced Interior Gateway Routing Protocol
- **OSPF:** Open Shortest Path First
- **BGP:** Border Gateway Protocol
- **VLAN:** Virtual Local Area Network
- **IEEE:** Institute of Electrical and Electronics Engineers
- **IP:** Internet Protocol